

Strategic Roadmap for Prioritizing of Predictive Maintenance Deployment in ceramic tiles factory in Algeria

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Abstract

The integration of advanced technologies, particularly Predictive Maintenance (PdM), is fundamentally transforming contemporary industrial maintenance strategies. This paper presents the initial phase of a comprehensive feasibility study investigating the deployment of PdM within the ceramic tile manufacturing sector in Algeria. The study pursues a dual objective : firstly, to identify the most critical machinery and production units to ensure a cost-optimized implementation of PdM ; and secondly, to maximize operational impact with minimized resource allocation. The overarching goal is to enhance the reliability and efficiency of ceramic tile production by leveraging advanced diagnostic tools to forecast equipment failures and proactively schedule maintenance interventions. The research methodology involves a thorough assessment of the current landscape and prevailing maintenance practices within the Algerian ceramic tile industry. This analysis is substantiated by data gathered through interviews conducted at more than six distinct manufacturing facilities. Additionally, the study evaluates the infrastructural prerequisites necessary to support a robust PdM framework. Preliminary results indicate significant potential for optimizing the production process through the adoption of PdM. An economic viability assessment, utilizing a Quality-affect analysis, confirms the project's feasibility and projects a positive return on investment. The study concludes that the targeted application of PdM to the most critical production units can reduce unplanned downtime attributable to equipment failure by over 40 %, thereby substantially bolstering the competitiveness and profitability of ceramic tile factories in Algeria.

Keywords : Industry 4.0 ; Strategic approach ; Predictive Maintenance ; Ceramic tiles factories, TOPSIS mixed 5M

1. Introduction

In an increasingly demanding and dynamic industrial context, the pursuit of higher productivity, efficiency, and competitiveness has become a strategic imperative for manufacturing sectors worldwide. One of the central challenges faced by industrial operators is the mitigation of unplanned equipment downtime, which can severely disrupt production flow, inflate maintenance costs, and compromise product quality. Within this framework, Predictive Maintenance (PdM) has emerged as a forward-looking solution to address these issues. PdM consists of anticipating failures before they occur through the integration of real-time data, intelligent sensors, and advanced analytics, thereby enabling interventions at the most opportune moment to prevent equipment breakdowns [1] [2]. The conceptual evolution of PdM has been largely driven by the rise of Industry 4.0 technologies. The deployment of the Internet of Things (IoT), machine learning algorithms, cloud computing, and digital twins has made it possible to collect, process, and analyze vast volumes of operational data in real time [3][4]. PdM is no longer limited to monitoring basic parameters such as temperature or vibration but increasingly relies on sophisticated prognostic models that detect degradation patterns and provide actionable insights [5][6]. Studies report that PdM can reduce unplanned downtime by 30 – 50%, extend equipment lifespan by 20 – 40%, and generate return-on-investment ratios ranging from 3 :1 to 10 :1 in certain industries [7][8]. While PdM has been widely adopted in high-tech and capital-intensive sectors such as aerospace, automotive, and energy, its diffusion into traditional manufacturing environments—especially in developing countries—remains relatively limited [9]. In Algeria, the ceramic tile industry represents a strategic sector in the national economy, characterized by high energy consumption, continuous production cycles, and strong sensitivity to process interruptions. However, maintenance practices in most Algerian ceramic factories remain largely preventive or corrective in nature, with limited integration of data-driven predictive techniques. This is partly due to organizational inertia, limited access to advanced technologies, and the lack of structured implementation methodologies. Recent feasibility studies conducted in the Algerian ceramic sector have identified the pressing unit as a critical area prone to

mechanical failures. While not explicitly labeled as a bottleneck, this zone plays a central role in the tile forming process and frequently experiences breakdowns that can significantly impact the entire production chain [10]. Targeting such bottleneck machines with PdM initiatives can yield disproportionately high benefits in terms of plant availability and throughput. Focusing maintenance efforts on bottlenecks is also aligned with the principles of the Theory of Constraints (TOC), which advocates for the prioritization of resources at the most limiting points in a system [11]. In addition to technical challenges, the successful implementation of PdM requires a strategic and organizational alignment. This includes a thorough analysis of production processes, identification of key assets, integration of sensing and monitoring infrastructure, and the training of personnel to interpret diagnostic outputs. A digital maturity assessment is also essential to gauge the plant's readiness to transition toward predictive maintenance [12]. Various frameworks, such as the IPSI (Implementation Phases for Smart Industry) model, propose staged approaches to PdM adoption, from data acquisition to decision support, tailored to the constraints of SMEs [13]. Moreover, economic evaluations must be considered early in the roadmap. Several publications emphasize the need for cost-benefit analyses, including metrics such as Payback Period (PBP), Net Present Value (NPV), and Return on Maintenance Investment (ROMI), to guide managerial decisions [14][15]. For small and medium-sized enterprises in Algeria, where investment capacities are limited, such analyses help justify the initial outlay required for sensors, data infrastructure, and human resource development.



Figure 1: Mindmap of this work

This article proposes a strategic approach for implementing Predictive Maintenance in Algerian ceramic tile factories, with a special focus on critical zone identification. This approach is structured in several stages : contextual analysis of the production system, identification and classification of critical equipment using 5M methods, and the prioritization of the zone of deployment. This methodology is particularly adapted to the Algerian context, where the integration of this type of maintenance in inexistent.

The originality of this study lies in its strategic approach, between root cause identification method and quality of product, while also considering the socio-technical conditions of local ceramic manufacturers. By grounding the

deployment of PdM in the operational realities of the factory and prioritizing interventions on critical equipments, this approach aims to maximize operational gains while minimizing technological disruptions. In doing so, it contributes to bridging the gap between theoretical advances in maintenance science and their practical application in emerging economies.

2. Literature Review

2.1. Predictive maintenance (PdM)

Historically, the simplest and most intuitive approach to industrial maintenance has been reactive, or corrective, maintenance. This strategy, often characterized as a "run-to-failure" model, involves taking action only after a piece of equipment has broken down or malfunctioned[16]. While this approach minimizes upfront planning and avoids the cost of servicing healthy equipment, its drawbacks are substantial, particularly in capital-intensive industries. Reactive maintenance is inherently disruptive, leading to unplanned downtime, which can halt entire production lines and result in significant financial losses from lost output and emergency repair costs [17]. This model is generally only suitable for non-critical, low-cost assets where the impact of a breakdown is minimal[18]. In complex, interdependent systems, such as those found in cement or glass manufacturing, the cascading effect of a single component failure makes a purely reactive strategy economically untenable and potentially hazardous [19]. The economic pressures and operational risks associated with unplanned downtime drove the adoption of a more structured, proactive approach : preventive, or planned, maintenance. This philosophy is based on performing maintenance tasks at predetermined intervals, which can be time-based (e.g., every six months) or usage-based (e.g., every 10,000 operating hours), irrespective of the asset's actual condition[16]. The primary goal of preventive maintenance is to reduce the frequency of failures and extend the overall lifecycle of assets by intervening before they are expected to fail based on historical averages and manufacturer recommendations[20]. While this approach represents a significant improvement over reactive maintenance by introducing predictability and reducing catastrophic failures, it is not without its own inefficiencies. Preventive schedules can lead to the premature replacement of components that are still in good working condition, thereby incurring unnecessary costs for parts and labor. Conversely, it cannot prevent failures that occur between scheduled service intervals, as it does not account for the specific operating context or degradation pattern of an individual asset [20]. The inherent limitations of both reactive and preventive strategies created the impetus for a more intelligent, data-driven paradigm. The shift towards truly proactive maintenance is fueled by the need to maximize the useful life of every component, enhance overall equipment effectiveness (OEE), and optimize resource allocation in a highly competitive global market [17]. This sets the stage for predictive maintenance, which seeks to resolve the inefficiencies of its predecessors by tailoring maintenance actions to the real-time health of each specific asset.

2.2. An Introduction to Maintenance Philosophies : From Correction to Prediction

Industrial maintenance has historically been governed by a reactive, or "run-to-failure," philosophy. In this model, corrective action is taken only after a component or system has failed. This approach, while simple to implement, is fraught with peril, leading to unscheduled and often catastrophic downtime, secondary damage to adjacent equipment, and significant safety hazards. The associated costs, both in terms of lost production and emergency repairs, are typically exorbitant. In an effort to mitigate these risks, industries widely adopted preventive maintenance. This strategy is predicated on performing maintenance at predetermined, fixed intervals—based on time, cycles, or mileage—regardless of the actual condition of the equipment. While representing an improvement over a purely reactive approach by preventing some failures, preventive maintenance is inherently inefficient. It often leads to over-maintenance, where perfectly healthy components are replaced prematurely, incurring unnecessary costs for labor and spare parts. Conversely, it can fail to prevent failures that occur between scheduled interventions, thus not fully eliminating the risk of unplanned downtime. The contemporary paradigm shift is towards Predictive Maintenance (PdM), a fundamentally different approach rooted in data and analytics. PdM is a proactive strategy that utilizes a data-driven methodology to analyze the real-time condition of equipment to predict when maintenance is required[17]. By continuously monitoring the health of an asset, PdM enables interventions to be scheduled with precision, just before a failure is likely to occur. This condition-based approach optimizes the allocation of maintenance resources, drastically reduces unplanned downtime, and maximizes the operational life of components[16]. The feasibility and scalability

of modern PdM are inextricably linked to the rise of Industry 4.0. The convergence of cyber-physical systems, the Industrial Internet of Things (IIoT), and advanced analytics has democratized the ability to collect and process the vast quantities of data required for accurate prediction. The proliferation of low-cost sensors and powerful computing platforms has transformed maintenance from a necessary cost center into a source of strategic competitive advantage, enabling smarter, more efficient, and more resilient manufacturing operations.

2.3. Comparative Analysis : Predictive vs. Preventive and Corrective Maintenance

The distinction between corrective, preventive, and predictive maintenance lies in their fundamental approach to timing, data utilization, cost structure, and overall impact on asset management. Corrective maintenance is event-driven, responding to failures after they occur. Preventive maintenance is schedule-driven, acting based on predetermined time or usage triggers. Predictive maintenance is condition-driven, initiating action based on data-informed forecasts of an asset's health.

The reliance on data is a key differentiator. While preventive maintenance may use historical, aggregated data like mean-time-between-failure (MTBF) to set schedules, PdM leverages a continuous stream of diverse, real-time data from the asset itself [21]. This allows PdM to account for the unique operational context and wear patterns of each machine, rather than treating all similar assets as if they degrade at the same average rate. Consequently, the financial implications differ significantly. Corrective maintenance is characterized by low upfront costs but can lead to extremely high, unpredictable expenses associated with emergency repairs and production losses [20]. Preventive maintenance involves regular, predictable operational costs, but these can be inflated by unnecessary servicing [20]. Predictive maintenance typically requires a significant upfront investment in technology (sensors, software, infrastructure) and expertise, but it aims to minimize the total cost of maintenance over the asset's lifecycle by optimizing both labor and component use while drastically reducing costly unplanned downtime [20]. According to some analyses, this approach can lead to a 35–50% reduction in downtime and a 20–40% increase in asset lifespan [21]. Table 1 provides a structured comparison of these three maintenance strategies across several key dimensions.

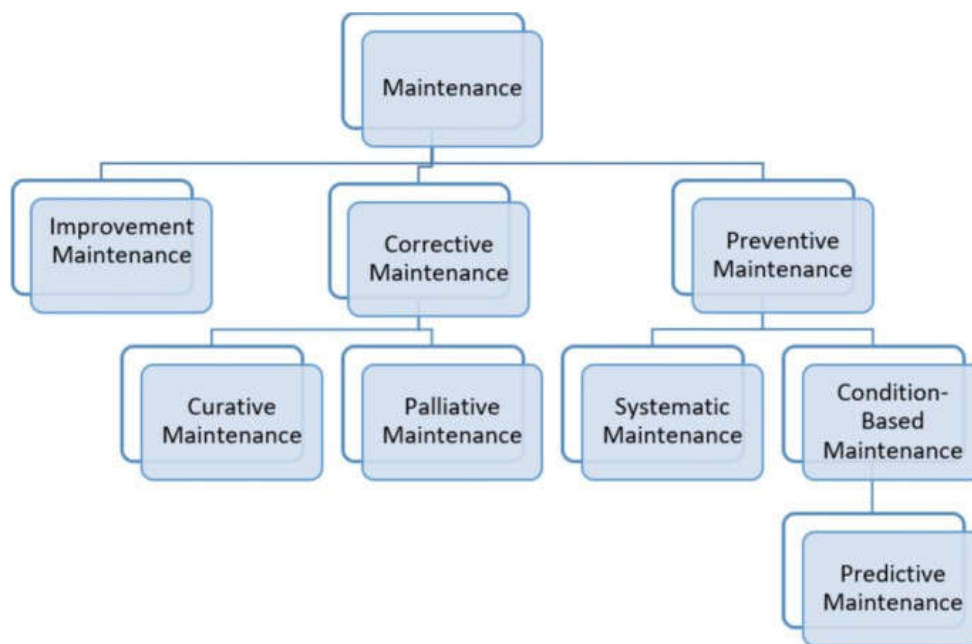


FIGURE 2: Types of Maintenance [22] [23]

2.4. The Paradigm Shift Towards Predictive Maintenance in Continuous Process Industries

The operational landscape of capital-intensive, continuous process industries such as cement, ceramics, and glass manufacturing is defined by a relentless pursuit of efficiency, reliability, and cost control. In this context, the ma-

Table 1: Comparative Analysis of Maintenance Strategies

Feature	Corrective Maintenance	Preventive Maintenance	Predictive Maintenance
Core Principle	Run-to-failure ; fix equipment after it breaks down. [16]	Time- or usage-based ; perform scheduled maintenance to prevent failures. [16]	Condition-based ; use data to predict failures and perform maintenance just-in-time. [16]
Timing of Intervention	Reactive : Occurs after failure. [20]	Proactive : Scheduled at regular, predetermined intervals. [20]	Proactive & Predictive : Triggered by the forecasted condition of the asset. [24]
Data Requirement	Event-driven ; failure reports and repair logs. [24]	Historical averages (e.g., MTBF) and manufacturer recommendations. [21]	Continuous real-time sensor data (vibration, temperature, etc.) and historical data. [25]
Cost Profile	Low upfront cost ; very high, unpredictable operational costs due to emergency repairs and downtime. [20]	Predictable, recurring operational costs ; potential for unnecessary spending on parts and labor. [20]	High upfront investment in technology ; lower long-term operational costs by optimizing maintenance and reducing downtime. [20]
Impact on Downtime	Results in significant, unplanned downtime. [20]	Reduces unplanned downtime but requires planned downtime for servicing. [21]	Aims to minimize both unplanned and planned downtime by optimizing maintenance intervals. [16]
Impact on Asset Lifespan	Can shorten asset life due to catastrophic failures and secondary damage. [20]	Extends asset life compared to corrective, but may involve premature component replacement. [20]	Maximizes asset lifespan by using components for their full useful life and preventing catastrophic failures. [16]
Typical Use Case	Non-critical, easily replaceable, low-cost assets. [18]	Safety-critical assets or equipment with predictable wear patterns. [18]	Critical, high-value, complex assets where downtime is extremely costly. [19]

nagement of physical assets is not merely a support function but a strategic imperative. Historically, maintenance philosophies have undergone a significant evolution, moving from reactive responses to proactive and, ultimately, predictive strategies. This paradigm shift, accelerated by the technological advancements of Industry 4.0, has repositioned maintenance as a critical driver of value, safety, and sustainability. This section traces this evolution, defines the core theoretical concepts underpinning modern predictive maintenance, and establishes the compelling economic and operational rationale for its adoption in high-value manufacturing environments.

3. Methodology

3.1. Analysis and observations

To refine our understanding of ceramic tile operations, we conducted semi-structured interviews with engineers, production and maintenance coordinators, line managers, and plant owners. Participants were purposively selected to capture technical, operational, and strategic viewpoints. Interviews were recorded, anonymized, and inductively coded ; Table 1 synthesizes recurrent themes by stakeholder role and frequency.

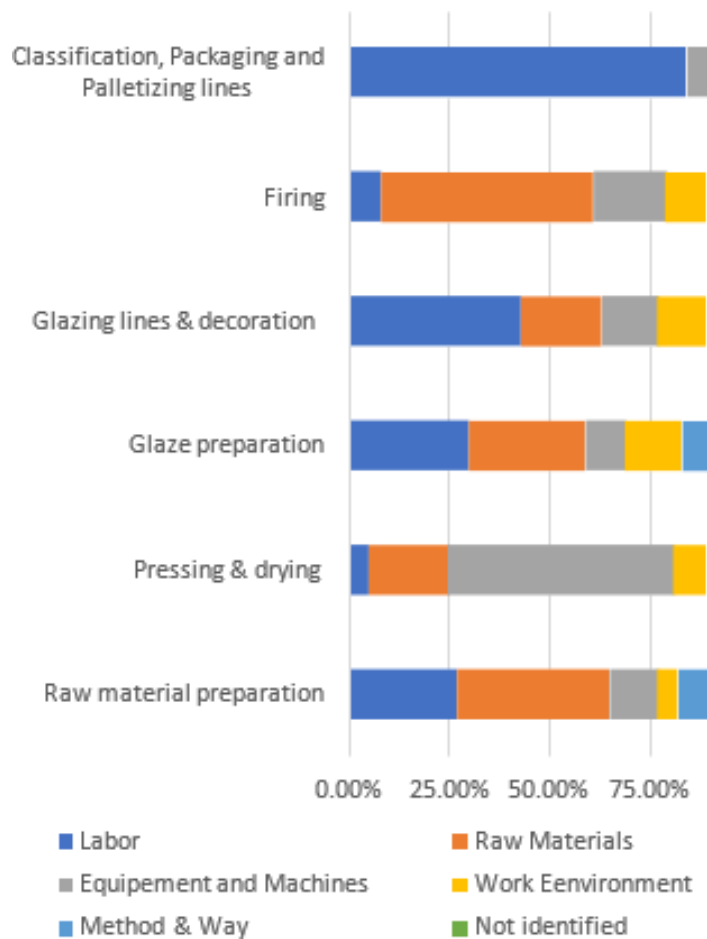


FIGURE 3: The percentages of defects cause according to their classification VS their units [10]

A consistent, context-specific finding concerns Algeria’s comparatively low industrial gas prices. Respondents emphasized that inexpensive energy enables genuine continuous operation, particularly stabilizing thermally sensitive firing and drying stages. Fewer start–stop cycles reduce scrap and thermal shock, sustaining steady-state quality.

Continuous running increases Overall Equipment Effectiveness by improving availability, smoothing throughput, and supporting uniform product characteristics.

This energy advantage enhances price competitiveness, strengthens schedule reliability, and sustains high capacity utilization under demand variability. It also broadens managerial options : kilns can remain hot during upstream interventions ; off-peak production can replenish safety stocks ; and predictive-maintenance windows can be aligned with energy-efficient baselines rather than costly restarts. Nevertheless, uninterrupted operation may accelerate wear of conveyors, kilns, and finishing equipment, underscoring the need for reliability-centered maintenance policies summarized in Table 2. recommendations.

Table 2: ANALYSIS AND OBSERVATIONS OF INTERVIEWS [10]

Answers	Analysis
A1	The production at this type of factory is continuous, i.e., 24h/7d
A2	Equipment and machines are mainly automated and semi-automated
A3	There are 6 dependent production units :
A4	The 6 units are important to varying degrees, the most important Firing and Pressing
A5	The majority of machine problems affect product quality
A6	Problems affect product quality are classified into 5 types according to Ishikawa 5M
A7	The production should never stop, the problems causing stoppages should be avoided
A8	There is no idea about PdM at some ceramic tile factory
A9	The factories produce from 5000 m ² up to 8500 m ² for each production line per day. (Varies from factory to factory)
A10	Sometimes suffering from sudden breakdowns
A11	Using EXCEL to manage our maintenance tasks and history
A12	There is no material to store hug data
A13	Spare parts available, but sometimes we suffer from rapture, we repair with worn parts
A14	The maintenance teams work day and night, only the managers work the days
A15	At the pressing and drying unit 56% of the problems are counted on the material
A16	The mean downtime : 30 min/day to 2 hours/day (varies from factory to factory)
A17	Type of maintenance tasks performed : routine check-ups, cleaning, repairs of the tile pressing machines, glazing machines and fix any issues that arise during production.
A18	The most important criterion is product quality and production continuity.

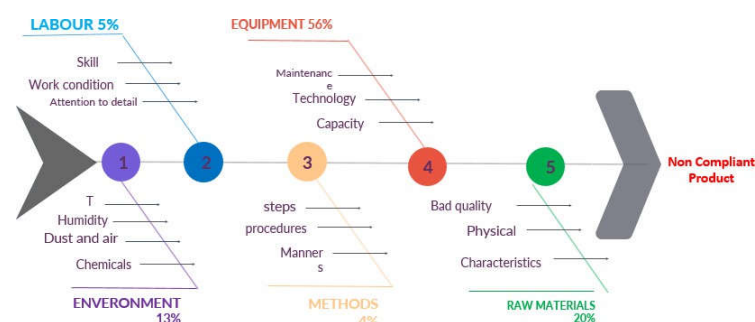


Figure 4: Ishikawa Classification problems at the pressing unit [10]

3.2. Classification problem due to quality product

The relatively low price of natural gas in Algeria enables ceramic tile plants to operate continuously while maintaining high energy efficiency. This structural cost advantage strengthens competitiveness and supports stable, sustained production output. Ceramic tile manufacturing is organized into six interdependent units. It begins with raw-material preparation—typically clay, sand, feldspar, and selected minerals—followed by body formulation in which constituents are blended to a plastic, dough-like consistency. The compound is then shaped into green tiles using a press or extruder and subsequently dried to controlled moisture levels. The dried bodies undergo kiln firing to remove residual moisture and densify the matrix, imparting mechanical strength. Finally, the tiles are glazed and subjected to a second firing to vitrify the coating, producing the desired smooth, often glossy surface.

3.3. Prioritization of PdM Deployment Across Workshops

We prioritize workshops for predictive maintenance (PdM) using the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS). Let $\mathbf{X} = [x_{ij}] \in \mathbb{R}^{m \times n}$ denote the decision matrix with $m = 6$ workshops and $n = 5$ criteria.

3.3.1. Step 0 : Data and Criterion Types

The decision matrix (percentages extracted from the figure) is :

Workshop	C_1	C_2	C_3	C_4	C_5
Raw material preparation	24.0	12.0	41.0	10.0	13.0
Pressing & drying	9.0	56.0	12.0	17.0	6.0
Glaze preparation	46.0	7.0	35.0	6.0	5.0
Glazing lines & decoration	41.0	8.0	24.0	22.0	5.0
Firing	12.0	9.0	52.0	23.0	4.0
Classification, Packaging and Palletizing lines	77.0	19.0	1.0	1.0	1.0

TABLE 3: Decision matrix $\mathbf{X}$ (percent).

Criterion types : C_2 is *Objective* ; C_1, C_3, C_4, C_5 are *Anti-Objective*.

3.3.2. Step 1 : Weight Determination (from empirical frequencies)

Weights are proportional to the column totals (frequency/percentage mass in the figure) and normalized to sum to 1 :

$$w_j = \frac{\sum_{i=1}^m x_{ij}}{\sum_{k=1}^n \sum_{i=1}^m x_{ik}}, \quad \sum_{j=1}^n w_j = 1.$$

Column sums are : $\sum_i C_1 = 209$, $\sum_i C_2 = 111$, $\sum_i C_3 = 165$, $\sum_i C_4 = 79$, $\sum_i C_5 = 34$. Hence the weights are :

Criterion	Weight w_j
C_1 Labor	0.349498
C_2 Equip. & Machines	0.185619
C_3 Raw Materials	0.275920
C_4 Work Environment	0.132107
C_5 Method & Way	0.056856

TABLE 4: Normalized weights from column-frequency information.

3.3.3. Step 2 : Vector Normalization

We compute the normalized matrix $\mathbf{R} = [r_{ij}]$ by the Euclidean norm per criterion :

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^m x_{ij}^2}}, \quad \text{''}\mathbf{x}\cdot\text{''} = \begin{bmatrix} x_{1j} \\ x_{2j} \\ \vdots \\ x_{mj} \end{bmatrix}.$$

(The norms are $\|\mathbf{x}\cdot_1\| = 102.6012$, $\|\mathbf{x}\cdot_2\| = 61.9274$, $\|\mathbf{x}\cdot_3\| = 79.5676$, $\|\mathbf{x}\cdot_4\| = 37.9342$, $\|\mathbf{x}\cdot_5\| = 16.4924$.)

Workshop	r_{i1}	r_{i2}	r_{i3}	r_{i4}	r_{i5}
Raw material preparation	0.2339	0.1938	0.5153	0.2636	0.7882
Pressing & drying	0.0877	0.9043	0.1508	0.4481	0.3638
Glaze preparation	0.4483	0.1130	0.4399	0.1582	0.3032
Glazing lines & decoration	0.3996	0.1292	0.3016	0.5800	0.3032
Firing	0.1170	0.1453	0.6535	0.6063	0.2425
Classification, Packaging and Palletizing lines	0.7505	0.3068	0.0126	0.0264	0.0606

Table 5: Normalized matrix $\mathbf{R}$.

3.3.4. Step 3 : Weighted Normalized Matrix

The weighted matrix is $v_{ij} = w_j r_{ij}$, i.e., $\mathbf{V} = \mathbf{R} \text{diag}(\mathbf{w})$:

Workshop	v_{i1}	v_{i2}	v_{i3}	v_{i4}	v_{i5}
Raw material preparation	0.0818	0.0360	0.1422	0.0348	0.0448
Pressing & drying	0.0307	0.1679	0.0416	0.0592	0.0207
Glaze preparation	0.1567	0.0210	0.1214	0.0209	0.0172
Glazing lines & decoration	0.1397	0.0240	0.0832	0.0766	0.0172
Firing	0.0409	0.0270	0.1803	0.0801	0.0138
Classification, Packaging and Palletizing lines	0.2623	0.0569	0.0035	0.0035	0.0034

Table 6: Weighted normalized matrix $\mathbf{V}$.

3.3.5. Step 4 : Ideal Best/Worst Solutions

For *benefit* criteria, $A_j^+ = \max_i v_{ij}$ and $A_j^- = \min_i v_{ij}$; for *cost* criteria, $A_j^+ = \min_i v_{ij}$ and $A_j^- = \max_i v_{ij}$. Given C_2 =benefit and $\{C_1, C_3, C_4, C_5\}$ =cost, we obtain :

$$\mathbf{A}^+ = [0.030657, 0.167852, 0.003468, 0.003483, 0.003447],$$

$$\mathbf{A}^- = [0.262291, 0.020982, 0.180323, 0.080098, 0.044816].$$

3.3.6. Step 5 : Separation Measures and Closeness Coefficients

The Euclidean separations are :

$$S_i^+ = \sqrt{\sum_{j=1}^n (v_{ij} - A_j^+)^2}, \quad S_i^- = \sqrt{\sum_{j=1}^n (v_{ij} - A_j^-)^2}.$$

The TOPSIS closeness coefficient is

$$C_i = \frac{S_i^-}{S_i^+ + S_i^-}, \quad 0 \leq C_i \leq 1,$$

and alternatives are ranked in descending order of C_i .

Workshop	S_i^+	S_i^-	C_i
Raw material preparation	0.2048	0.1906	0.4820
Pressing & drying	0.0697	0.3090	0.8160
Glaze preparation	0.2277	0.1374	0.3764
Glazing lines & decoration	0.2109	0.1589	0.4297
Firing	0.2392	0.2237	0.4832
Classification, Packaging and Palletizing lines	0.2568	0.2004	0.4383

TABLE 7: Distances to ideal best/worst and closeness coefficients.

3.4. Identification of the most critical unit

Sorting by C_i (descending) gives the following prioritization :

1. **Pressing & drying** ($C = 0.8160$)
2. **Firing** ($C = 0.4832$)
3. **Raw material preparation** ($C = 0.4820$)
4. **Classification, Packaging and Palletizing lines** ($C = 0.4383$)
5. **Glazing lines & decoration** ($C = 0.4297$)
6. **Glaze preparation** ($C = 0.3764$)

Interpretation.. Because C_2 (Equipment & Machines) is a benefit criterion with a substantial weight and very high relative value for *Pressing & drying*, this workshop lies closest to the ideal solution. Cost criteria (C_1, C_3, C_4, C_5) penalize high percentages ; workshops with lower burden on these dimensions and/or strong performance on C_2 rise in priority. The mid-tier group is separated by small differences in C (e.g., Firing vs. Raw material preparation), which can be further refined if you introduce additional criteria or plant-specific constraints (e.g., safety criticality, cost-to-implement, sensing readiness).

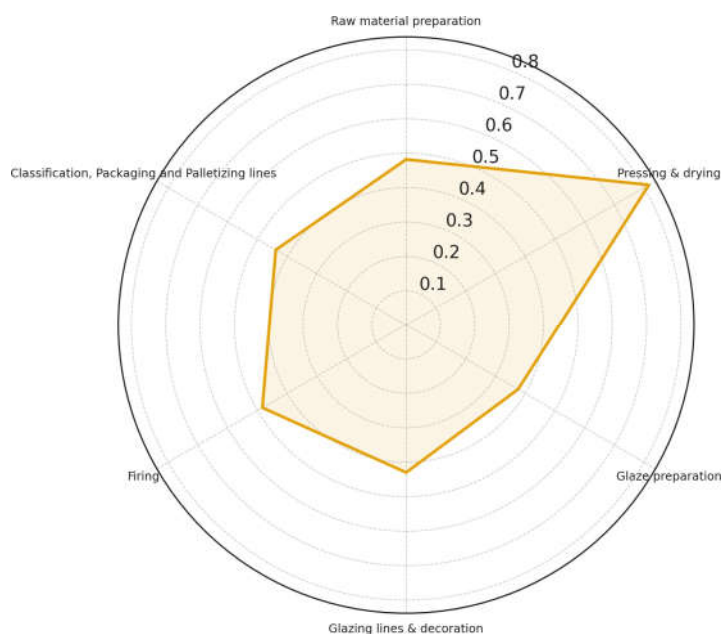


FIGURE 5: TOPSIS Closeness Coefficient per Workshop/Unit

4. Discussion and Interpretation of Results

The primary objective of this study was to develop a data-driven strategic roadmap for deploying Predictive Maintenance (PdM) within the Algerian ceramic tile industry. Recognizing the operational vulnerabilities inherent in prevailing maintenance strategies, the research employed a hybrid methodology, integrating 5M (Ishikawa) root-cause analysis with the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS). This approach aimed to optimize resource allocation by identifying the production units where PdM implementation would yield the most significant impact.

4.1. 1. Interpretation of the TOPSIS Prioritization

The application of TOPSIS successfully translated the empirical defect data (classified via the 5M framework across six workshops) into a quantitative ranking. The results unequivocally identify the Pressing & Drying unit as the highest priority for PdM deployment, achieving a closeness coefficient (C_i) of 0.8160. This score significantly outpaces the second-ranked unit, Firing ($C_i=0.4832$). The dominance of the Pressing & Drying unit is driven primarily by the high incidence of defects attributed to “Equipment and Machines” (C2) within this specific workshop (56%). A crucial aspect of interpreting these results lies in how the criteria were defined within the TOPSIS model. The authors designated C2 (Equipment) as a “benefit” criterion, while the others (Labor, Raw Materials, Environment, Method) were designated as “cost” criteria. In this context, “benefit” signifies the opportunity or potential advantage gained by deploying PdM. Since PdM technologies are specifically designed to mitigate equipment failures, a unit exhibiting the highest rate of such failures represents the most fertile ground for realizing the advantages of PdM. This prioritization correctly targets the area where the proposed intervention will have the maximum impact, aligning with the principles of the Theory of Constraints (TOC) by focusing resources on a critical bottleneck.

4.2. 2. Strategic Implications of the 5M Analysis

The underlying 5M analysis reveals distinct operational profiles for each workshop, which informs the overall maintenance strategy and demonstrates that a “one-size-fits-all” approach to PdM is inefficient.

- **Equipment-Centric Failures** : The Pressing & Drying unit is unique as equipment is the dominant failure driver. This confirms its suitability as the primary target for advanced diagnostic tools.
- **Labor-Dominant Areas** : Conversely, the Classification, Packaging, and Palletizing lines show that 77% of defects are attributable to Labor (C1). While this area experiences significant issues, PdM is not the optimal solution ; investments in automation, training, or process optimization would yield better results.
- **Diversified Profiles** : Units such as Firing and Raw Material Preparation (the second tier of priority) are heavily impacted by Raw Materials (C3) and Labor (C1), alongside equipment concerns. PdM is relevant here, but the immediate gains are projected to be less impactful than in the Pressing & Drying unit, requiring a strategy that also addresses material consistency and workforce practices.

4.3. 3. Contextual Factors and Feasibility

The results must also be viewed within the specific context of the Algerian ceramic industry. The study notes that low industrial gas prices enable continuous (24/7) operation. While this stabilizes thermally sensitive processes like firing, it concurrently accelerates the wear of mechanical components, particularly in the Pressing & Drying unit. This operational reality further underscores the urgency of transitioning to a predictive approach to manage this accelerated degradation and avoid costly disruptions. Furthermore, the interviews (Table 2) highlighted existing challenges, such as a lack of awareness regarding PdM (A8), reliance on basic tools like Excel for maintenance management (A11), and insufficient infrastructure for data storage (A12). The success of the roadmap depends not only on identifying the correct unit but also on addressing these foundational infrastructural and cultural deficits.

4.4. 4. Limitations

While the methodology is robust, certain limitations should be acknowledged. The TOPSIS weights were determined based on the empirical frequencies of reported issues. Sensitivity analysis using alternative weighting methods (e.g., expert judgment via Analytic Hierarchy Process) could further validate the stability of the prioritization. Additionally, the study aggregates data from multiple factories ; facility-specific constraints might slightly alter the p

5. Conclusion

This study establishes a strategic roadmap for the prioritized deployment of Predictive Maintenance (PdM) within the Algerian ceramic tile manufacturing sector. Recognizing the operational vulnerabilities inherent in prevailing corrective and preventive maintenance strategies in this continuous-process industry, the research aims to optimize resource allocation by identifying the most critical production units. A hybrid methodology, integrating the 5M root-cause analysis with the TOPSIS multi-criteria decision-making technique, is employed to systematically prioritize workshops according to their impact on product quality and operational continuity.

The analysis, substantiated by empirical data from several Algerian facilities, decisively identifies the *Pressing & Drying* unit as the highest priority for PdM intervention. This unit exhibits the greatest susceptibility to equipment-related failures (56% of issues), making it the optimal focal point for initial deployment. The targeted application of PdM in this critical area is projected to substantially enhance reliability and efficiency, with assessments suggesting a potential reduction in unplanned downtime exceeding 40%.

Looking forward, future research should focus on the practical implementation phases of this roadmap, specifically the selection of appropriate sensor technologies and the development of machine-learning models tailored to the failure modes of *Pressing & Drying* equipment. Longitudinal studies are necessary to validate the projected efficiency gains and to assess the scalability of this framework to other units, such as *Firing* and *Raw material preparation*. Furthermore, investigating socio-technical challenges—including workforce upskilling and organizational change management—will be crucial for ensuring the sustainable adoption of Industry 4.0 technologies in this context.

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